

**CATHOLIC JUNIOR COLLEGE**
JC2 PRELIMINARY EXAMINATION
Higher 2**MARKING**
SCHEME**BIOLOGY****Paper 3 Long Structured and Free-response Questions****9744/03****13 September 2021****2 hours**

Candidates answer on the Question Paper.

Additional Materials: Writing Papers and Appendix A

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

Section AAnswer **all** questions in the spaces provided on the Question paper.**Section B**Answer **one** question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten your answers for **Section B** securely.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [30]	
2 [9]	
3 [11]	
4 or 5 [25]	
TOTAL	75

1. Reason 1: (idea of) The LNPs protect the mRNA from being digested by nucleases, reacting with water, oxygen or disrupted due to change in pH;
2. Explanation 1: (idea) mRNA is single stranded; hence less stable / mRNA has OH group at 2' end; hence relatively reactive.
3. Reason 2: (idea of) The LNPs Allow the mRNA to be uptake by the cells
4. Explanation 1: mRNA is negatively-charged due to its phosphate group, unable to pass through cell surface membrane.

Fig 1.2 shows some organelles in a muscle cell.



Fig. 1.2

- (b) With reference to Fig. 1.2, name **two** organelles other than mitochondria that are involved the synthesis and secretion of the SARS-CoV-2 S glycoprotein from the muscle cell after the uptake of comirnaty into the cell. Justify your answer.

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Any of the 2 naming and corresponding justification:

1. Naming 1: Rough endoplasmic reticulum

2. Justification 1: (idea of) ribosomes on the RER serve as site for translation and lumen of RER allows the polypeptide to undergo folding, glycosylation and to be transported.
3. Naming 2: Golgi body / apparatus
4. Justification 2: (idea of) Golgi body / apparatus serves as site for the S glycoprotein to be further modified, sorted and packaged into secretory vesicles
5. Naming 3: ribosomes
6. Justification 3: (idea of) ribosomes serve as site for translation
7. Naming 4: nucleus
8. Justification 4: (idea of) nucleus contains DNA that codes for rRNA, ribosomal proteins and tRNA and tRNA synthetase that are required for translation.

(c) Explain how the termination phase of translation allows the newly synthesised polypeptide to be released from the translation assembly.

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1. A **stop codon** on the **mRNA** reaches the **A-sites** of the **ribosome** where **A-site** accepts a protein factor called a **release factor** instead of another aminoacyl-tRNA (**accept, simply tRNA**)
2. The release factor causes the **addition of water** to the **growing polypeptide chain**;
3. This leads to **hydrolysis** of the **polypeptide chain** from the **tRNA** it is bound to.

(d) Explain how mitochondria is involved in the synthesis of the polypeptide of S glycoprotein in the muscle cells that took in the mRNA-LNPs.

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1. Mitochondria is a **site** for **ATP synthesis** (reject: energy)
2. **Activation of amino acid / attachment of amino acid to tRNA / formation of amino-acyl tRNA** catalysed by **aminoacyl-tRNA synthetase** for the translation requires **ATP**.

A study was performed to investigate the effect of different concentrations of oxygen (O_2) on the synthesis of **S** glycoprotein in muscle cells that took in the mRNA-LNPs. All other conditions are kept constant.

Under normal condition, the concentration of O_2 in the air is around 20%. In addition, the normal concentration of lactic acid in muscle cells is around 0.50 – 2.00 mM. Beyond 4.00 mM of lactic acid, cell death occurs.

Table 1.1 shows the result obtained for the study.

Table 1.1

Concentration of O_2 / %	Average amount of S glycoprotein obtained from 5.0×10^6 muscle cells after 24 hours / μg	Average concentration of lactic acid obtained from 5.0×10^6 muscle cells after 24 hours / mM
20	0.410	0.50
15	0.382	1.80
10	0.251	3.62
5	0.093	4.50
0	0.021	4.50

(e) With reference to Table 1.1,

- (i) account for the effect of the decrease in concentration of O_2 from 20% to 15% on the synthesis of **S** glycoprotein in the muscle cells.

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1. **Description:** As the concentration of O_2 decreases from 20% to 15%, the amount of **S** glycoprotein synthesised **decreases slightly** from 0.410 to 0.382 μg .

Explanation:

2. As the concentration of O_2 decreases, the rate of aerobic respiration decreases. Less ATP synthesised and to be used for translation.
 3. (idea of) The decrease in ATP synthesis is compensated with the increase in rate of lactate fermentation / production for regeneration of NAD^+ ;
 4. allowing for ATP synthesis via substrate level phosphorylation during glycolysis.
 5. (idea of) less ATP synthesised during anaerobic respiration compared to aerobic respiration; therefore the compensation is not sufficient to produce the same amount of ATP to sustain the same rate of translation.
- (ii) suggest why muscle cells subjected to 5% and 0% of O_2 were still able to produced small amount of **S** glycoprotein despite the high concentration of lactic acid that is beyond 4.00 mM.

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1. (idea of) The data shown in Table 1.1 is based on 24 hours incubation time, the cells were able to synthesised **S** glycoprotein using mainly the ATP synthesised through glycolysis
2. (idea of) As the time progressed, the amount of lactic acid produced and accumulated is too high that it leads to cell death. Hence, only small amount of **S** glycoprotein was synthesised.

Fig 1.3 shows the uptake of mRNA-LNPs into cell.

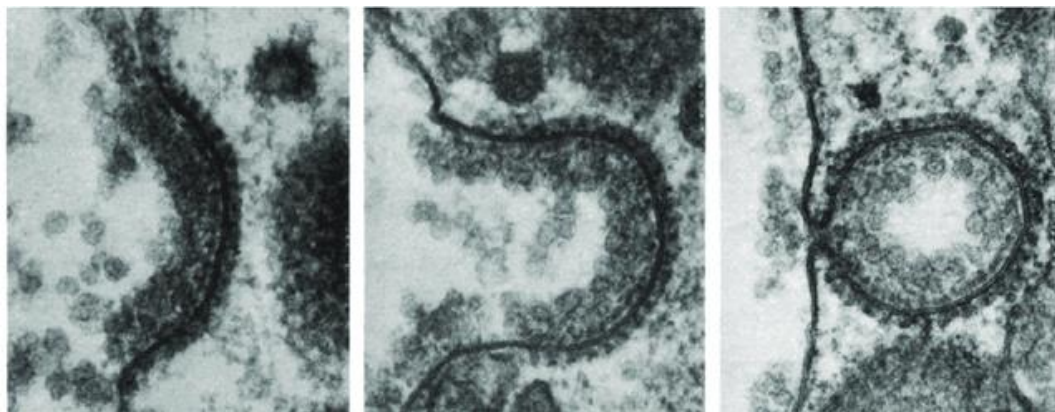


Fig. 1.3

- (f) A student claimed that mRNA-LNPs do not enter cells through fusion with cell surface membrane. Instead, mRNA-LNPs enter cells through endocytosis. Based on the information in Fig. 1.3; justify the claim.

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1. (idea of) In **fusion**, the mRNA-LNPs membrane becomes contiguous with the cell surface membrane.
2. In **endocytosis**, the cell internalises the mRNA-LNPs by wrapping it in an endosomal vesicle / (idea of) invagination is present as shown in Fig. 1.1.

(g) The structure of mRNA used in comirnaty vaccine resembles the structure of mRNA found in the cytoplasm of eukaryotic cells.

Explain how the structure of mRNA in comirnaty vaccine allows it to serve its role inside the muscle cells.

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1. (idea of) **5' methylguanosine cap** allows the mRNA to bind to / interact with the **ribosome** for the **initiation of translation**.
2. (idea of) Codons on the mRNA allows the mRNA to serve as the **template** for synthesis of **S** glycoprotein.
3. (idea of) **Poly-A tail** protects the mRNA so that the codon / coding sequence of the mRNA is not digested by the **nucleases** [reject: enzyme] present in the cytoplasm.

The effectiveness of comirnaty vaccine was studied in two groups of population residing in region **A** and region **B** respectively. All other factors were kept constant between the two groups except for the region where they are residing.

Blood sample was taken from 5 individuals from each regions to measure the concentration of antibody against **S** glycoprotein of SARS-CoV-2.

Table 2.2 below shows the data collected from the study.

Table 2.2

Place of residence	Concentration of antibody against S glycoprotein of SARS-CoV-2 / mM / 100 mL of blood
Region A	5.2
	6.7
	5.8
	6.9
	5.5
Region B	6.2
	5.4
	4.8
	4.3
	6.3

- (h) Using an appropriate statistical test, determine if the difference in the effectiveness of comirnaty in people residing in region **A** and region **B** is statistically significant.

Show your working in the space below.

(You are provided with **Appendix A** which contains the required statistical formulae, *t*-test Distribution Table and χ^2 -test Distribution Table to solve this question)

Null hypothesis :

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Alternative hypothesis :

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[1] – Correct Null hypothesis and Alternative hypothesis

[1] – Correct calculation of mean and standard deviation for effectiveness of vaccine for people residing in region A and B

Mean for Region A = 6.02

Mean for Region B = 5.4

Standard deviation for Region A = 0.746324326

Standard deviation for Region B = 0.868907359

[1] – Correct calculation of *t*-test ^{calculated} value

$$t_{\text{calculated}} = (6.02 - 5.4) / 0.512249938$$

$$= 1.210346656$$

[1] – Correct identification of *t*-test ^{critical} value

$$t_{\text{critical}} = 1.859548 \text{ (at degree of freedom = 8, } p = 0.05)$$

[1] – Correct conclusion

[5]

[Total: 30]

- 2 Genetically identical bacteria cells; *Escherichia coli* (*E.coli*) were divided equally into two groups; namely:

Control Group : the *E. coli* cells continue to be grown under optimum conditions.

Treatment Group : the *E. coli* cells continue to be grown under the same optimum conditions as in the control group. In addition, this group of *E. coli* cells are incubated with λ bacteriophages carrying ampicillin (a type of antibiotic) resistance gene (*AmpR*).

After 24 hours of incubation, the *E. coli* cells from the **Treatment Group** was found to have acquired resistance to ampicillin while the cells from the **Control Group** was lacking the resistance to ampicillin.

- (a) With reference to the information above, name the horizontal gene transfer that occurred in the **Treatment Group**. Explain your answer.

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 [2]

1. **Transduction** [accept either: generalised transduction or specialised transduction]
2. The process involves **lambda bacteriophage**.

- (b) Without the use of molecular techniques, suggest how the presence of *AmpR* gene in the **Treatment Group** could be inferred.

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 [2]

1. Grow / culture the *E.coli* from the **Treatment Group** and **Control Group** on medium / environment containing the antibiotic: **ampicillin**.
2. If the **Treatment Group** survives / grow but the **Control Group** cannot survive / grow, this suggests the presence of *AmpR* gene in the **Treatment Group**.

A separate experiment was performed on the **Control Group** and **Treatment Group** mentioned above after the 24 hours of incubation.

This experiment aims to investigate the effect of the horizontal gene transfer process on the amount of chromosomal DNA in the *E. coli*.

Fig. 2.1 shows the result of the experiment.

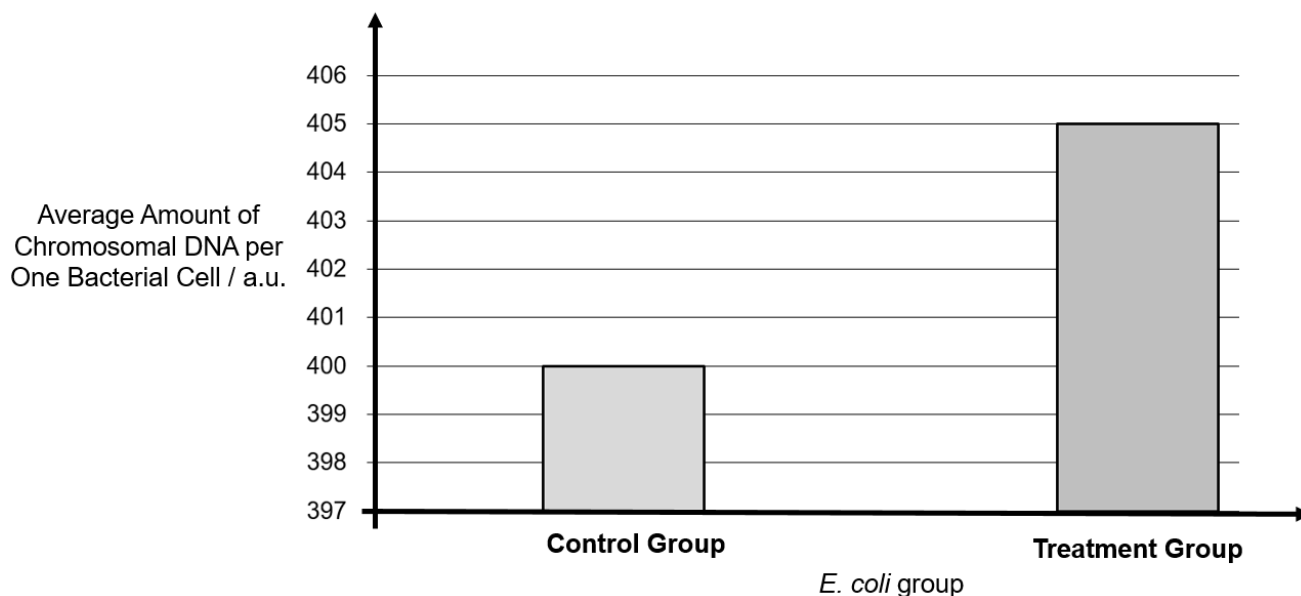


Fig. 2.1

Table 2.2 provides information on the size of various DNA.

Table 2. 2

DNA	Size in kb
<i>E.coli</i> chromosome	4,000
λ phage genome	48.8
<i>AmpR</i> gene	1.2

- (c) With reference to Fig. 2.1 and Table 2.2, account for the change in the average amount of chromosomal DNA per one bacterial cell in the **Treatment Group**.

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- The average amount of chromosomal DNA per one bacterial cell in the **Treatment Group** **increases by 5 a.u.** as compared to the **Control Group**.
Explanation:

2. the λ phage transferred its entire genome and the *AmpR* gene into the *E.coli* in the Treatment Group and integrated them into the bacterial chromosome.
3. Comparing Fig. 2.1 and Table 2.2, DNA of a size 4,000 kb gives rise to approximately 400 a.u of DNA as seen in the result for the Control Group. 1 kb of DNA = 0.1 a.u of DNA.
4. λ phage genome + *AmpR* gene = 50 kb gives rise to approximately 50 kb x 0.1 a.u. = 5 a.u. of DNA which accounts for the 5 a.u. increase in the amount of DNA in the Treatment Group.

- (d) The scientists would like to amplify the *AmpR* gene using Polymerase Chain Reaction (PCR). However, the DNA sequence of *AmpR* gene is unavailable.

Suggest why PCR could not be used to amplify the *AmpR* gene.

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 [1]

1. DNA sequences flanking the *AmpR* gene are not known, the primers flanking the *AmpR* gene cannot be synthesised.

[Total: 9]

- 3 Dengue fever is caused by the dengue virus (DENV), of which there are 4 serotypes DENV1 – DENV4. The disease is transmitted when an infected *Aedes aegypti* mosquito bites a human, injecting the virus. *Aedes aegypti* has rapidly expanded its range in the last 50 years.

Where once it was confined to the tropics, this mosquito has been found at high altitudes and higher latitudes. Global warming is now driving this mosquito out of its normal range and into new, uncharted latitudes. Along with the mosquito, DENV is now cropping up in countries that never recorded a single dengue fever case.

- (a) Explain how global warming has caused the spread of DENV beyond the tropics.

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 [2]

1. Global warming has caused an increase in temperature and humidity around the world, moving the temperature barrier of countries above the tropics towards the poles.
2. DENV's vector, *Aedes aegypti*, can now survive and colonise habitats it could not before.

Public health efforts to counter the rise in dengue transmission during the wet season is to frequently clean out rain gutters and empty out containers and pots.

- (b) Describe the conditions *Aedes aegypti* requires for propagation.

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 [1]

1. The female mosquito needs **stagnant water and warm temperatures** for egg laying.

In tropical countries where *Aedes aegypti*'s population is already well established, small increases in overall temperature can benefit the mosquito's physiology and behaviour.

- (c) Explain how climate change affects insects' physiology.

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1. Insects are ectothermic, hence their **internal temperature depends on the external temperature**.
2. As temperature increases, insects' **enzymes gain more kinetic energy**, and are able to form **more enzyme-substrate complexes faster**.
3. This would increase rate of reaction of metabolic activities, and insect larvae would **develop faster into adults/ egg hatching time may reduce/size of insects may increase due to increased metabolism & growth**.

With countries where more than one serotype of DENV is circulating, citizens are vulnerable to infections from all the 4 serotypes (DENV1 – DENV4) equally, regardless if they had been infected by different serotype before. It seems that being infected by one serotype does not give immunity against other serotypes

- (d) Explain why an individuals are not immune to other serotypes after being infected with one serotype.

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1. exposure to 1 DENV serotype triggers adaptive immune response that results in production of **memory B cells with BCR specific to Ag of this serotype** (conferring immunological memory);
2. however, the **4 DENV serotypes have slightly different antigen / surface protein structure** / antigenicity;
3. **BCR of memory B cells are not fully complementary** to Ag of DENV of a different serotype;
4. thus subsequent exposure to **different DENV serotype will trigger primary immune response** rather than a secondary response;

Max 3 points

- (e) Describe briefly the immune response that is responsible for the fever symptom that sets in almost immediately once a patient gets infected with dengue.

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[2]

1. Innate immune response, **mast cells secrete histamine** that causes **blood vessel dilation and increased permeability**,
2. for antimicrobial peptides and **macrophages/neutrophils to migrate** to the site of infection that secrete **more cytokines, causing increased blood flow** to the infection site, causing heat and the onset of a fever.

[Total: 11]

Section B

Answer **one** question in this section

Write your answers on separate answer paper provided.

Answer each part on a **separate** piece of paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a) Photosynthesis is a complex process, with reference to the structure and function of chloroplast, explain how the efficiency of photosynthesis is maximized in plants. [13]

1. In the Light dependent reaction, **harvesting of light energy** is maximized with the **use of different pigments e.g. carotenoids / chlorophyll a / chlorophyll b** which utilize different wavelengths of light
2. Energy captured is **channeled to photosystems** so that efficiency of light energy harvested is maximized. OWTTE.
3. **Photosynthetic Pigments** involved in photophosphorylation are **arranged on the thylakoid membrane 90° / perpendicular to the direction of sunlight**. OWTTE.
4. Increases the efficiency of physical capture of Light energy. OWTTE.
5. **Folding of the thylakoid membrane** allows for an **increase in surface area**
6. Which allows for **more pigments to be packed on the thylakoid membrane** and efficiency maximized.
Regarding the replenishment of electrons to photosystems
7. The **arrangement of cytochromes** involved in photophosphorylation **allows for a reduction in energy** from PSII to the ETC and finally to PSI. OWTTE.
8. This allows for the **electron to be of an energy low enough so it can be accepted by PSI** and replenish PSI in the process. OWTTE
9. Structural coupling of **PSII** with **Photolysis of water** allows for
10. The electron **replenishment of PSII** with the splitting of water and evolving of oxygen as a byproduct.
ATP Synthesis

11. Cytochromes in the **Electron transport Chain / ETC** are **coupled with proton pumps**. OWTTE.
12. **Free energy** that is **lost due to the transfer of electrons** from one cytochrome to the next is **used by proton pumps** which actively pump protons from the Stroma to the Thylakoid space.
13. **Folding of the thylakoid membrane increases the surface area** of the membrane.
14. This allows for a **greater number of stalked particles** each containing ATP synthase which allows for **maximizing of ATP synthesis**.
15. **Structural positioning of photosynthetic pigments** that allow for **cyclic photophosphorylation** to take place.
16. Cyclic photophosphorylation in turn **facilitates the generation of additional ATP** needed so that the **Light independent reaction can continue**. OWTTE.
Light independent reaction
17. Structurally the **thylakoid membrane** and the **inner chloroplast membrane**, sees to the **compartmentalization of processes**, / **creates a common region (Stroma)**. OWTTE.
18. within which multiple copies of the **enzymes involved in the Calvin cycle are held in common** which sees to the **regeneration of Ribulose Biphosphate (RuBP)** in a **faster time** along with the **synthesis of Glucose**. OWTTE.

- (b) There are many factors that contribute to evolution in eukaryotes. Explain the roles of chance and isolation in an evolution that leads to the emergence of a new species. [12]

In evolution chance plays a significant role,

1. In the **Struggle for Survival**, conditions that deplete or **reduce the population** are greatly left to chance, e.g. **disease / degree of predation / competition for food / habitat** etc. OWTTE.
2. This would allow for a **Constant population** to be established which affords a greater chance for evolution depending on the severity of the selection pressure. OWTTE.

In terms of Variation within the gene pool

3. Chance also plays a part in the **accumulation of Single Base Pair Substitutions / Point Mutations** that allow a **gain of function**.
4. Chance also plays a part in the **occurrence of Single Base Pair deletions / additions / Point Mutations** that render a **loss of function**.
5. Chance also plays a part in the **translocation of a gene to a highly expressed promoter / chromosomal aberrations** that allow a gain of function. OWTTE.
6. Chance also plays a part in the **gene addition / duplication resulting in a higher gene expression / chromosomal aberration**. OWTTE.
7. Chance also plays a part in **aneuploidy / Numerical aberrations** that leads to additional gene expression. OWTTE.
8. Chance also plays a part in **polyploidy / Numerical aberrations** that leads to additional gene expression. OWTTE.
9. Chance also plays a part in **independent assortment and random segregation during Meiosis** that leads to different combinations within gametes. OWTTE.
10. Chance also plays a part in **random fusion of gametes** that leads to different combinations within the zygote. OWTTE.
11. Chance also plays a part in **random mating** that leads to different combinations within the zygote. OWTTE.

In Natural Selection

12. Chance also plays a part in having **sufficient time** for accumulation of mutations before a specific selection pressure is exerted. OWTTE.
13. Chance plays a part in the gene being **present in the gene pool** at the time of selection.

In the mode of inheritance / production of similar offspring

14. Chance also plays a role in **heterozygous protection** in **preservation of the recessive allele in the gene pool**.
 15. Chance in the **combination of alleles** due to **Linkage or Codominance** also contributes to the selection and subsequent survival of the population.
- MAX 6**

Isolation is important in evolution in the following ways:

16. **Geographical isolation** which **restricts gene flow** allowing a population to gain distinction over time from another population. OWTTE.
17. This will lead to **allopatric speciation** over time. OWTTE.
18. **Temporal isolation** also **restricts gene flow** allowing a population to gain distinction over time from another population. OWTTE.
19. **Behavioural isolation** also **restricts gene flow** allowing a population to gain distinction over time from another population. OWTTE.
20. **Mechanical isolation** also **restricts gene flow** allowing a population to gain distinction over time from another population. OWTTE.
21. **Post zygotic mechanisms** also allow a population to gain distinction over time from another population. OWTTE.
22. **Isolation** allows a **population to have distinction** from another population over time.
23. Where there is **no geographical isolation**, all other forms of isolation also lead to **sympatric speciation** over time. OWTTE.
24. Describing how each level of isolation is achieved.

Max 6

[Total: 25]

- 5(a)** "Respiration is a series of oxidation and reduction processes." Discuss the purpose of oxidation and reduction during respiration. [13]
Source: Novel

Glycolysis

1. During glycolysis, **glucose is oxidised** during the **double dehydrogenation of glyceraldehyde-3-phosphate (GALP) into pyruvate**.
2. The **purpose** of this dehydrogenation is to **generate 2 molecules of electron carriers NADH for oxidative phosphorylation/electron transport chain**. / **2 molecules of NAD⁺ are reduced to produce NADH which while enter OP**.
3. **2 molecules of ATP via substrate level phosphorylation** so that the cell has some ATP during anaerobic respiration.

Link Reaction

4. Oxidative decarboxylation occurs where **pyruvate is oxidised into acetyl CoA**.
5. The hydrogen released is **accepted by NADH** to again be **channelled into oxidative phosphorylation**. / **NAD⁺ is reduced to NADH to be channelled into OP**.
6. The **purpose** of the link reaction is to convert pyruvate into a molecule that **can enter the mitochondrion**.

Krebs Cycle

7. Krebs cycle is the **complete oxidation of acetyl CoA** into CO_2 & H_2O .
8. **3 NAD^+ and 1 FAD^+ are reduced** during the various steps of the Krebs cycle to **provide electrons for the ETC/oxidative phosphorylation**.
9. **Citrate is oxidised/ dehydrogenated** and decarboxylated into **α -ketoglutarate and one molecule of NAD^+ is reduced into NADH** here
10. **α -ketoglutarate is further oxidised/dehydrogenated** into succinate, reducing **one molecule of NAD^+ in the process**.
11. **Succinate is oxidised/dehydrogenated** into fumarate, **reducing FAD^+ into FADH_2** in the process.
12. **Malate is oxidised/dehydrogenated** into oxaloacetate, reducing **one final molecule of NAD^+ into NADH**.

Oxidative Phosphorylation

13. During oxidative phosphorylation, electron carriers **NADH & FADH_2 are oxidised** when they **donate electrons to the electron transport chain**.
14. Electrons are then passed down the ETC, **reducing each subsequent electron carrier that accepts it / oxidising the electron carrier that it leaves**.
15. As electrons flow down the ETC, **they lose a little energy each time** which is **used by hydrogen carriers to pump H^+ into the intermembrane space**.
16. The H^+ gradient/proton gradient created generates a proton motive force, which causes H^+ to diffuse down ATP synthase. ATP synthase uses H^+ to phosphorylate ADP to ATP.
17. **Oxygen is reduced** as it acts at the final electron acceptor.
18. All **NADH and FADH_2 coming from glycolysis, link reaction and Krebs cycle** are able to stimulate the **production of 28 ATP** molecules via oxidative phosphorylation.
19. **QWC – for an answer that is organized and makes repeated connections to the importance oxidation/reduction plays in each step of respiration**.

- 5(b)** In the human population, different combination of phenotypes are observed in each individual, except identical twins.

Explain how the different combination of phenotypes in each individual could have arisen. [12]

Source: Novel

1. The **phenotype** of an organism is the **physical or chemical characteristic that is expressed**.
2. Phenotypes are **determined by the genotype/genes of the organism** which refers to the genetic constitution/allele combination the organism has for a particular characteristic.
3. Depending on the nucleotide sequence, **there can be more than one allele for a particular gene with each different allele** coding for a slightly different phenotype.
4. Each individual will inherit two copies of a gene, one from each parent. The **resulting phenotype is dependent on which allele is expressed**.

5. Allele expression can be **influenced by the combinations that come via meiosis, Mendelian interaction, co-dominance, incomplete dominance, linkage and epistasis.**

Meiosis

6. During **prophase I** of meiosis, **crossing over of non-sister chromatids of homologous chromosomes** may occur which results in exchange of alleles.
7. During **metaphase I**, alleles of different characteristics on different genes **assort independently** from each other.
8. Thirdly, during **anaphase I**, **homologous chromosomes segregate independently** from each other.
9. These processes all allow for **different combinations of dominant and recessive alleles for each characteristic to be found in gametes** which would thus give organisms a large amount of phenotypes.

Mendelian

10. Genes that follow mendelian inheritance would **have two alleles that can code for a particular characteristic.**
11. The **allele that is expressed when the individual is heterozygous** for the gene is known as the dominant allele, while
12. The **recessive phenotype is expressed only if the individual is homozygous** for the recessive allele.

Co-dominance/ & Incomplete Dominance

13. **Co-dominant alleles would be both equally expressed in a heterozygote**, with the phenotype showing both versions of the characteristic.
14. **Incomplete dominance** would see the **heterozygote expressing a phenotype that is an intermediate of** the phenotype of an individual that is homozygous for either alleles.

Linkage

15. Genes found on the **same chromosome would be inherited together and would not independently assort.**
16. This would **influence the phenotypes that arise after the organism has developed** as certain traits tend to be inherited together eg: hair colour and eye colour.
17. Genes that are **located on sex-chromosomes would be linked to the gender** of the offspring, and phenotypes arising from these genes would show a preference to a particular gender. OWTTE.
18. **Sex-linked diseases that are caused by recessive alleles** tend to be **more common amongst males** as they would only carry a single copy of the gene compared to females.

Epistasis

19. **Epistasis** is where two genes are required for a characteristic to be expressed, and the **allele of one gene masks the phenotypic expression of a second gene at an entirely different locus.**

[Total: 25]